

PRESSURE MEASUREMENT FILM PRESCALE

**SUPER LOW PRESSURE
LLW
(TWO-SHEET TYPE)**

① LINE UP

Nine types of Prescale are supplied according to Eight pressure level. Select the appropriate Prescale range.

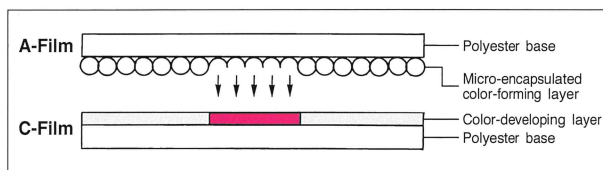
type	Film type	Pressure range[MPa] $1\text{MPa} \div 10.2\text{kgf/cm}^2$ Pressure range[psi] $1\text{psi} \div 6895\text{Pa}$									
		0.006	0.05	0.2	0.5	0.6	2.5	10	50	130	300
		0.87	7.3	29	73	87	363	1450	7250	18850	43500
Two-sheet type	Ultra Extreme Low Pressure (5LW)	[Bar]									
	Extreme Low Pressure (4LW)	[Bar]									
	Ultra Super Low Pressure (LLLW)	[Bar]									
	Super Low Pressure (LLW)	[Bar]									
	Low Pressure (LW)	[Bar]									
Mono-sheet type	Medium Pressure (MW)	[Bar]									
	Medium Pressure (MS)	[Bar]									
	High Pressure (HS)	[Bar]									
	Super High Pressure (HHS)	[Bar]									

* Film type W:Two-sheet
S:Mono-sheet

② STRUCTURE AND HOW IT WORKS

Structure

There are two types of Prescale; Two-sheet type and Mono-sheet type.
Two-sheet type is composed of two polyester bases. One is coated with a layer of micro-encapsulated color forming material (A-film) and the other with a layer of the color developing material (C-film). Use two films facing the coated sides each other.



How it works

When pressure is applied, the microcapsules are broken and the color-forming material reacts with the color-developing material to make red color. The microcapsules are designed to break according to the pressure so the color density corresponds to the pressure.

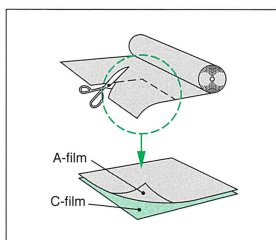
③ PROPERTIES

Precision	$\pm 10\%$ or less (measured by densitometer at 23°C/73.4°F, 65% RH)
Recommended temperature range	20°C~35°C(68°F~95°F)
Recommended humidity range	35% RH~80% RH

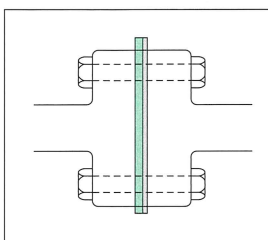
④ HOW TO USE

Two-sheet type(Ultra super low~Medium pressure: LLLW~MW)

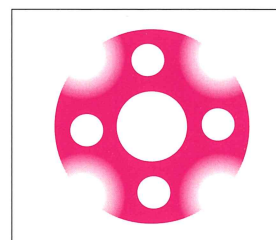
Cut the two films appropriately. (A-film in a black poly sack and C-film in a blue poly sack) Face the rough surfaces of each films and insert the films where you want to measure pressure. Apply pressure. Red patches appear on the film and the color density changes according to pressure level. Take out the C-film, see and check the pressure distribution.



(1) Cut the Prescale Film into the required shape. With the Two-sheet film, make sure the coated sides on A-Film and C-Film face each other.



(2) Insert cut Prescale Film into area to be measured and apply pressure.



(3) Remove film and observe pressure distribution.

SUPER LOW PRESSURE

LLW

(TWO-SHEET TYPE)

Measurement pressure range: 0.5–2.5MPa

• Pressure application conditions

Time to reach the pressure to be measured: 2 min.

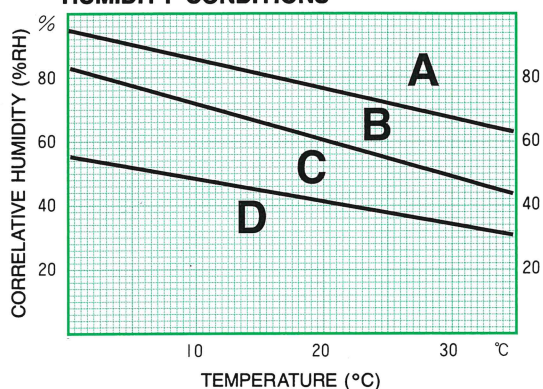
Time of retention at the pressure to be measured: 2 min.

Check if the temperature and humidity meet with the conditions above when the pressure is applied.

(For example, if the room temperature is 25°C and the humidity factor is 60%RH, acquire the pressure from the B curve in the standard chart.)

STANDARD CONTINUOUS PRESSURE CHART

GRAPH OF TEMPERATURE/
HUMIDITY CONDITIONS



STANDARD COLOR SAMPLE

1.5



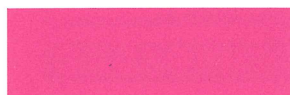
1.3



1.1



0.9



0.7



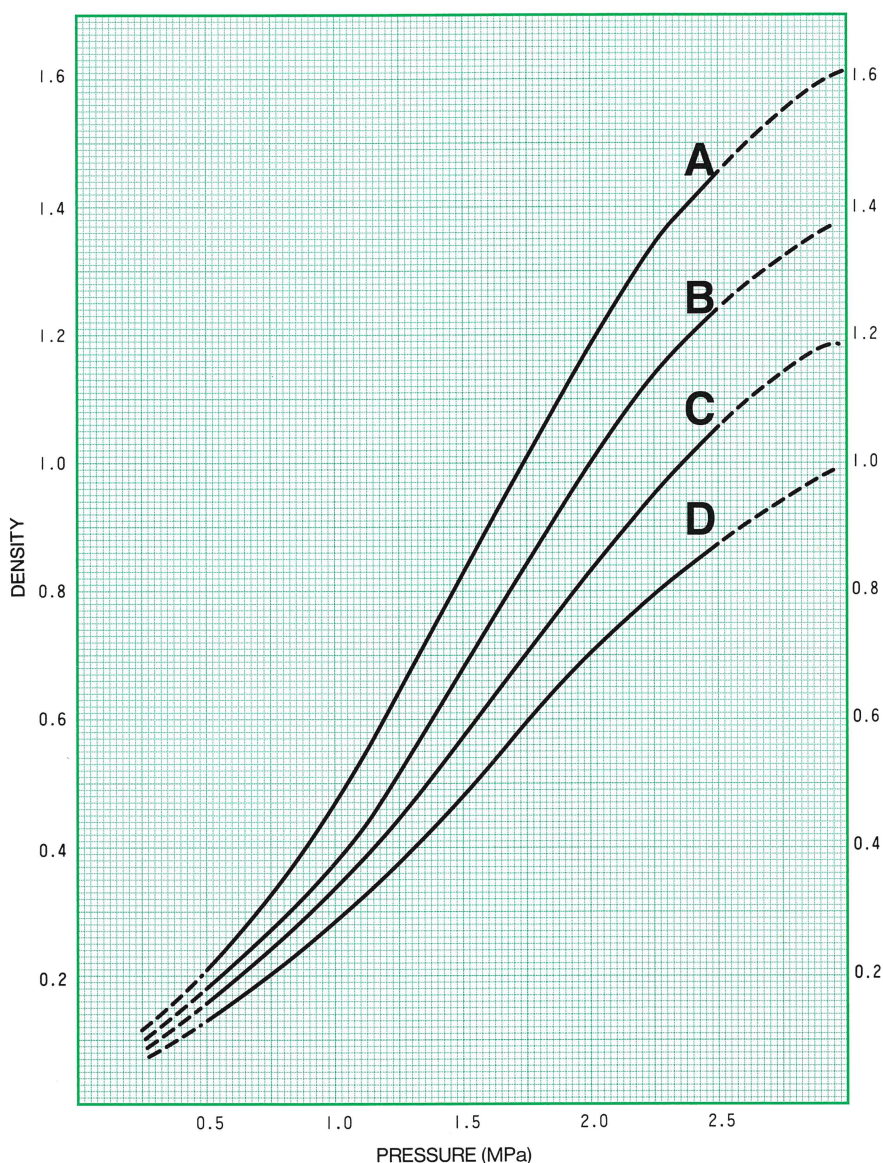
0.5



0.3



0.1



As the pressure range indicated by the broken line in the graph may exceed the permissible error range, it should be used for reference purposes only.